

Time Machine on macOS Ventura (2017+ Hardware) with RAID Mirroring and Cloud Backup: Performance & Benefits

Overview of Time Machine on Modern macOS

Apple's **Time Machine** is the built-in backup system in macOS, designed to work seamlessly with local drives (USB, Thunderbolt, etc.) and even network storage ¹. Introduced back in 2007 (OS X 10.5), Time Machine has evolved significantly. Notably, macOS Big Sur and later (including **Ventura**) revamped Time Machine to use the APFS file system and snapshot technology, enabling *"faster, more compact, and more reliable backups"* than the old HFS+ format ². Independent tests found that the new APFS-based Time Machine can be roughly 2.7× faster for the initial backup and 4× faster for subsequent backups, compared to the previous version ³. This is especially relevant for Macs from **2017 or newer**, which are fully compatible with these software improvements. In practice, Time Machine is commonly used with an external drive connected to a Mac (via USB-C, Thunderbolt, etc.), such as an iMac 2017 paired with a high-capacity external disk ¹.

Time Machine's core idea is simple: it performs **incremental, versioned backups** of your system automatically. Once you turn it on, it will **back up all files on your Mac that are not part of macOS itself** (this means user applications, documents, photos, music, source code, etc.) ⁴ ⁵. Backups occur **hourly**, with macOS automatically managing a **rotation schedule** (hourly snapshots for 24 hours, daily snapshots for the past month, and weekly snapshots for older data) ⁶. These backups allow you to restore the entire system or individual files to earlier versions as needed. Time Machine even integrates with apps (like Photos or Mail) to restore items within those apps' interface ⁷. All of this happens in the background with minimal user intervention.

How Time Machine Works (Ventura and Later)

Time Machine in Ventura uses APFS **snapshots** and incremental copy-on-write logic to efficiently store changes. The first time you back up to a new disk, Time Machine copies everything (this is the longest backup). Thereafter, each hourly backup copies **only the changes since the last backup**, which is much faster and smaller in size ⁸. For example, if you add or modify 500 MB of files in an hour, only that 500 MB (plus some overhead) will be written to the backup during the next run. This design allows continuous data protection without saturating your storage after the initial backup. Apple notes that after your first full backup, *"future backups should be faster"* precisely because only changes are saved ⁹.

Local Snapshots: On APFS-formatted Macs, Time Machine also leverages local snapshots. macOS will save an hourly snapshot of the internal disk (if space allows) and keep it for up to 24 hours ¹⁰. This means if your external backup drive is temporarily disconnected (e.g. a laptop on the go), you still have 24 hours of version history stored on your Mac's own drive. Once you reconnect your Time Machine disk, those local

snapshots get merged into the backup. Local snapshots are automatically managed (deleted as needed for space) and are available only on APFS volumes ¹¹ .

Backup Content and Exclusions: By default, Time Machine covers **almost everything** in your user space – documents, research data, applications you’ve installed, etc. ⁴ . It explicitly **excludes the macOS system files** and apps that came with the OS installation ⁵ (since those can be restored by reinstalling macOS, and excluding them saves space and time). You can also manually exclude specific files or folders via Time Machine settings if needed (more on this below). Time Machine keeps a **history** of file versions, so you can recover an earlier copy of a file or a deleted file by “entering” the Time Machine interface and navigating the timeline of backups ¹² . This versioning is extremely useful for scientific and AI work – for instance, you can retrieve last week’s version of a dataset or source code if a new change introduced errors.

APFS Requirement: One important technical note – since macOS 11 Big Sur, Time Machine **requires the backup disk to be APFS-formatted** (if you set up a new backup). If you connect an old HFS+ Time Machine disk, macOS will prompt to reformat it. In fact, if you use a RAID or certain managed drives, you must format them as APFS for Time Machine to work on Big Sur/Ventura ¹³ . Users found that if a disk wasn’t APFS, Time Machine would get stuck preparing the backup until it was reformatted to APFS ¹⁴ . The switch to APFS was worth it: Apple reports significantly improved performance and reliability, and as noted earlier, tests confirm major speed gains on APFS backups ² . For example, one user using a Thunderbolt 3 RAID array reported that after reformatting to APFS, an **initial 750 GB backup completed in about 1.5 hours** ¹⁵ – an impressive throughput on the order of 140 MB/s. This shows that modern hardware (like a 2017 iMac with Thunderbolt or USB 3) can achieve high backup speeds, especially for the first backup, when the backup target and connection are not bottlenecks.

Performance Considerations

Background Throttling: It’s important to understand that Time Machine is designed as a *background service* and macOS will **throttle its resource usage** to avoid impacting your work. By default, Time Machine backup processes are set to a low I/O priority (Apple classifies it as “Tier 3” or `IOPOL_THROTTLE` priority) – meaning the system lets higher-priority tasks (like your apps) have disk and CPU priority ¹⁶ . In practical terms, this throttling inserts small delays when disk contention is detected, especially on slower drives. *How does this affect performance?* During the first full backup, macOS knows it’s a big one-time job and tends to allow more use of performance resources (P-cores, faster I/O) ¹⁷ . Measurements have shown initial backups often run much faster than subsequent hourly backups. For instance, one benchmark saw a fast SSD write at **900–1100 MB/s** during the first backup, but then the routine hourly backups (with throttling in effect) ran at only **190–365 MB/s** for sequential writes and as low as 3–32 MB/s for lots of small files ¹⁸ . In other words, the system deliberately *slows down* Time Machine when it’s running automatically, so that it doesn’t overwhelm other tasks. This is by design and usually not a cause for concern – backups still complete in a reasonable time, and the Mac remains responsive.

HDD vs SSD Performance: If your backup drive is a mechanical hard disk (HDD) – as is likely with a large 20 TB RAID – the throttling has a somewhat bigger impact. Apple uses different throttle timings for SSDs versus HDDs. Specifically, when delaying backup I/O, the OS will pause writes longer on a hard disk (sleeping ~200 ms) than on an SSD (~25 ms) in each cycle ¹⁹ ²⁰ . This means backups to spinning disks are slowed more aggressively to avoid saturating the drive. Additionally, APFS itself can be **slower on HDDs** due to fragmentation of metadata over time ²¹ . The key point: a large, spinning external drive (like two 10 TB drives in RAID1) will not back up as fast as an SSD, especially after long usage. However, given typical data

workflows, this is usually fine – Time Machine will still get the job done, just not at the maximum raw interface speed. It's also worth noting that the first backup is often unthrottled or less throttled (especially if started manually), so you might observe high speeds initially, and then more moderate speeds for subsequent hourly increments ¹⁷ .

Impact on a 2017 iMac: The iMac (2017) 5K is an Intel-based Mac that can run Ventura and handle Time Machine well. It likely has either a Fusion Drive or SSD internally; either way, it can push high data rates over Thunderbolt 3 or USB 3. For instance, a USB 3.0 connection to a single HDD can sustain ~100–150 MB/s, and Thunderbolt to a RAID can be higher. With Time Machine's throttling, the actual observed transfer might be lower, but it will utilize idle resources efficiently. The **goal of Time Machine is not to finish the backup as fast as technically possible, but to finish it without interfering with your usage** ²² . In a scientific computing context, this means you can be training models or crunching data while Time Machine quietly copies changes in the background. If the workload is very I/O intensive (e.g. you are streaming data to disk at high rates), Time Machine will politely yield and slow its pace.

Large Files and Frequent Changes: Performance is also influenced by the *nature of your data*. Time Machine works at the file level for backups (it uses file system events and snapshots to determine what changed). If you have very **large files that change frequently**, each time they change Time Machine will copy the *whole file* again in the next backup ²³ . For example, imagine a 5 GB database or a virtual machine disk image that updates daily – every backup will include a fresh 5 GB copy if any portion of that file changed. This can obviously impact backup duration and storage. In AI research, you might have large datasets or model checkpoints that update often. Apple specifically calls out that “*unusually large files that update frequently*” can make backups “*time-consuming*,” and recommends considering excluding them from backups if appropriate ²³ . We'll discuss strategies for that in a later section. The APFS snapshot approach slightly mitigates this by copying only changed blocks into snapshots (making incremental differences more space-efficient) ²⁴ , but over time even changed blocks of a large file can accumulate to effectively the whole file size ²⁵ .

Initial Backup Experience: It's worth highlighting that the **first Time Machine backup** is the only time you'll move all data at once. This can take anywhere from minutes to hours depending on how much data you have. In your case, if your iMac has, say, 1–2 TB of data and it's going to a 20 TB external RAID over USB 3 or Thunderbolt, the first backup might indeed run for several hours (for 2 TB it could be on the order of 4–6 hours over USB 3, assuming ~150 MB/s effective throughput). This is usually an overnight job. The example earlier (750 GB in 1.5 hrs) was on a very fast Thunderbolt RAID ¹⁵ – your mileage may vary. After that initial pass, though, the hourly backups of a typical workday's changes (maybe a few GB) will likely complete in a few minutes or less each. Time Machine also **skips running if the Mac is asleep** or off; it will simply catch up when the Mac is on again (it doesn't “queue” missed hours beyond saving local snapshots). On desktops like iMac, which are often always on, it will run hourly like clockwork. If an hourly backup is still running when the next hour comes, macOS will not start a second instance – it will let the current backup finish (they effectively run serially). So if you ever *do* push massive amounts of new data continuously (imagine adding 100 GB every hour), Time Machine might end up running nearly constantly to keep up. In such extreme cases, you might opt to adjust backup frequency or exclude certain data, but for most workflows this isn't a problem.

RAID Mirrored Storage as a Backup Destination

Now let's consider the storage setup: a **20 TB My Book Duo configured as RAID 1 (mirroring)**, used for Time Machine. In RAID 1, the two drives inside the enclosure are exact copies of each other – all data is written to both simultaneously, creating a mirrored set. This does *not* increase capacity (20 TB mirror = 10 TB usable, since the second drive is essentially a duplicate), nor does it significantly increase write speed (writes go to both drives, so you're limited by the speed of one drive). The main benefit is **redundancy**: if one of the two drives fails, your Time Machine backup is still intact on the other drive. This is a great addition to data safety, because an un-mirrored external drive is a single point of failure (drives *do* fail, eventually). With a mirror, the chance of both drives failing at the same time is very low. You can replace the failed unit and rebuild the mirror without losing the backup. This setup aligns with the principle of keeping **multiple copies** of data on different media to protect against failure ²⁶ – in fact, within the external unit you already have two copies of the backup data (one on each disk).

A few notes on using RAID with Time Machine on macOS Ventura:

- **Formatting & Compatibility:** The My Book Duo likely presents itself as a single drive to the Mac (hardware RAID). You would format that single volume as APFS (preferably APFS Encrypted). Apple's official documentation on Time Machine doesn't explicitly mention RAIDs, but it treats them like any other external drive – as long as the Mac can mount it, Time Machine can use it ²⁷. We saw earlier that if using a software-managed RAID like SoftRAID, you *must* use APFS format or Time Machine won't work ¹³. With a hardware RAID enclosure, just ensure it's an APFS volume. Ventura's Time Machine preference will typically prompt to erase and reformat a disk to APFS if it's not already in the correct format ²⁸ ²⁹.
- **Performance of RAID 1:** Because mirror writes data twice, there may be a slight overhead – but practically, writing 100 MB/s to two drives vs one drive isn't a big difference since the limiting factor is likely the USB interface or drive speed. Some RAID enclosures can read from both drives in parallel (improving read speed), but Time Machine's usage pattern (writing backups and occasionally reading during restores) won't greatly benefit from faster reads except during a full restore. In short, using RAID 1 should feel about the same as using a single drive for backup in terms of speed. The crucial difference is reliability. (If one drive fails, macOS might not even "know" if the enclosure handles it transparently – the enclosure would ideally notify you via a utility or LED. It's good to periodically check the RAID status with the WD utility, if available, to ensure both disks are healthy.)
- **Disk Failure and Rebuilds:** In a mirror setup, when a drive fails, you replace it and the RAID hardware rebuilds the new drive from the remaining good drive. This rebuilding process will churn the disks (lots of reading/writing for hours) – it might be wise to pause Time Machine during a rebuild to avoid stressing the only good copy or slowing the rebuild. Once the mirror is restored, Time Machine can resume normal operation, continuing to keep both drives identical. Keep in mind that while RAID 1 protects from drive hardware failure, it doesn't protect against other risks like *accidental deletion* (if you delete a file, that deletion is mirrored) or *corruption* (if the filesystem gets corrupted, it's mirrored too). That's why RAID is not a substitute for having **separate** backups – it's a high-availability feature, not a true backup by itself. In our case, the RAID is an extra layer on *top* of the backup.

- **Encryption:** You mentioned using **encrypted Time Machine backups**, which is highly recommended. When you set up Time Machine and enable encryption, the external APFS volume is encrypted with AES (XTS-AES 128, typically). On a 2017 iMac (which lacks the Apple T2 security chip), encryption is done in software, but thanks to CPU acceleration (Intel chips have AES-NI), the performance hit is modest. Apple notes that APFS encryption works at the volume level, and it doesn't double-encrypt if a drive is already encrypted – enabling “APFS Encrypted” essentially means the entire backup filesystem is encrypted on disk ³⁰ ³¹ . The peace of mind is worth it: if someone steals that external drive, they cannot read your research data or personal files without the encryption password. Just be sure you remember the password or have it stored in your iCloud Keychain (macOS usually offers to remember it for you). Losing access to an encrypted backup due to a forgotten password is a disaster to avoid. In summary, with APFS you get strong encryption with little friction – definitely a good practice since you handle sensitive or valuable data.
- **Backup Disk as “Read-Only”:** A quick aside – in APFS Time Machine, the backup volume is mounted as a read-only volume in normal use. You'll notice in Finder that you can browse “Backups” but not modify them. This is intentional to prevent tampering. Time Machine manages the data via snapshots. Just something to be aware of: don't try to manually store extra files on the Time Machine disk; Apple's documentation advises using the disk **only** for Time Machine to avoid any interference ³² .

Including Cloud Backups (iCloud, Dropbox, etc.) for Off-Site Safety

While Time Machine (especially on a RAID1 drive) gives you an excellent on-site backup, it's wise to also have **off-site backups** – this is where cloud services come in. In the backup strategy world, there's a rule of thumb called the **3-2-1 rule**: keep **3** copies of your data, on at least **2** different types of storage media, with at least **1** copy kept **off-site** (away from the others) ²⁶ . In our scenario, you have the original data on your iMac (internal drive) and a local backup on the external RAID (that actually has two mirrored copies, but both are local). To fully follow 3-2-1, you'd want another copy in a different location – cloud storage fulfills that role nicely.

iCloud: Apple's iCloud is built into macOS and provides a form of cloud backup/sync for specific data. If you sign in with an Apple ID and enable iCloud features, it can automatically sync your **Photos**, contacts, calendars, notes, passwords, and so on, and can also sync files via **iCloud Drive** ³³ . For example, if you enable “Desktop & Documents” in iCloud Drive, those folders will be mirrored to Apple's cloud. iCloud's advantage is that it's seamless and keeps data *accessible anywhere*, not just backed up ³³ . Indeed, Apple markets iCloud as keeping your data “*safe, automatically backed up, and available anywhere*” ³³ . However, it's important to clarify that **iCloud is primarily a synchronization service, not a full Mac backup**. It doesn't make a historical archive of your whole system in the way Time Machine does. If you delete a file from an iCloud-synced folder, it gets removed from iCloud too (though there is a recently deleted recovery period for iCloud Drive files). Also, iCloud only covers what you choose to sync: for many users that's photos, documents, and app data – which might actually be the most critical subset of their data.

One thing to watch out for: if you use the “**Optimize Mac Storage**” feature, your Mac will offload some infrequently used iCloud files from your local disk (to save space), keeping them only in the cloud. Time Machine **cannot back up files that aren't on the local disk**. So if Optimize Storage is on and, say, a large folder is only stored in iCloud (with a download-on-demand icon), Time Machine will *not* include that data in its backup ³⁴ . In essence, iCloud is already storing it, so you still have a copy – but it's something to be

mindful of. If you want everything to be both in iCloud and in Time Machine, you might disable optimization so that all files stay downloaded locally (requiring enough disk space).

Despite these nuances, **iCloud is very useful as a tertiary backup** for key data. It provides **off-site coverage** – if (heaven forbid) your entire office burned down, the Time Machine drive and Mac might both be lost, but your iCloud data (documents, photos, etc.) would still exist on Apple’s servers. It’s also continuously updating (as long as you have internet), which complements Time Machine’s hourly schedule.

Third-Party Cloud (Dropbox, Google Drive, etc.): Services like Dropbox, Google Drive, OneDrive, or Box operate similarly to iCloud Drive in that they sync selected folders/files to the cloud. Many researchers use Dropbox or Google Drive to easily share data with colleagues and maintain an off-site copy. The same caveats apply: these are **sync services**. They typically have version history for files (e.g., Dropbox keeps 30 days of versions on personal accounts), which is somewhat backup-like, but they are not intended to snapshot your whole system. You choose what folders to put in the sync folder. For example, you might keep your active project data in a “Dropbox” folder so that every edit is sent to the cloud immediately. This is great for collaboration and off-site safety of that project. It doesn’t automatically cover, say, your entire home directory unless you manually put everything in there (which isn’t practical for something like the whole Photos library perhaps).

One way to leverage these: identify **critical subsets** of your data – perhaps your source code repositories, papers and notes, or processed results – and ensure those live in a cloud-synced folder (or at least are copied to one regularly). That way you have an *additional* backup beyond Time Machine. For instance, you might use iCloud Drive for your Desktop/Documents, and use Dropbox or Google Drive for specific research data that you want redundant on another platform or to share with collaborators. There’s no harm in using multiple services if you have them.

Dedicated Cloud Backups: Beyond sync services, there are true **cloud backup services** like Backblaze, CrashPlan, or Carbonite, as well as tools like Arq that let you back up your Mac to cloud storage (Amazon S3, Google Cloud, etc.). These act more like Time Machine in the cloud – automatically backing up everything (with version history), regardless of folder, and keeping it in an encrypted archive online. The Apple Support site even acknowledges that *“some [third-party utilities] can back up to a cloud-based service”* and you should use whatever meets your needs ³⁵. The advantage of these is comprehensive coverage and set-and-forget off-site backup. The downsides are cost (for large amounts of data, you’ll pay ongoing fees) and the initial upload can be slow for huge datasets (your 2 TB could take weeks to upload on a typical internet connection). In your case, if you have tens of terabytes of AI data, uploading it all might be impractical, so you’d prioritize what truly needs an off-site copy. Some researchers keep an **offline clone** that they periodically update and store off-site (e.g. a removable hard drive in a safe deposit box) as an alternative – but using cloud is far more convenient if the data size is manageable.

Combining Strategies (Local + Cloud): The good news is that local Time Machine and cloud backups don’t conflict – they address different threat models. Time Machine covers you for quick recovery, versioning, and mistakes on a day-to-day basis; cloud/off-site covers you for catastrophic loss (and also allows access to some files when away from the Mac). A well-rounded approach is to use **both**. In fact, experts strongly encourage leveraging both local and cloud solutions, since each has strengths: local disks are cheap and fast, while cloud is robust and off-site – using both compensates for the weaknesses of each ³⁶. Following this philosophy, you achieve redundancy *and* geographic diversity. For instance, Cloudwards (a backup review site) states: *“Hard drives are cheap and fast, but lack reliability for long term storage. Although the cloud*

is far more reliable, its cost and slow transfers mean it's not a silver bullet. For a well-rounded solution, follow the 3-2-1 rule and use a combination of the two." ³⁶ ³⁷ . In your case, you already have the combination in mind: Time Machine on a redundant local disk (RAID1) plus iCloud/Dropbox. This effectively implements 3-2-1: you have (at least) 3 copies of important data (original, local TM, cloud), on 2 different media (local disk and cloud servers), and 1 off-site copy (cloud).

A brief mention on HomeKit/Apple hardware integration (which you noted you **do not** want to focus on): Time Machine is mostly independent of things like HomeKit. There isn't really an integration there – e.g., you can't trigger Time Machine via Siri or HomeKit automations in any meaningful way. It runs on its schedule or on-demand when you click "Back Up Now." Likewise, the specific Mac hardware (whether it's a Mac mini or iMac or MacBook) doesn't change how Time Machine operates, beyond performance differences. So we won't dwell on hardware gadgetry; the focus remains on software behavior and backup strategy.

Backup Coverage and Data Management for Research/AI Work

With a **20 TB** Time Machine volume (10 TB usable in mirror) dedicated to backups, you have a very large space to hold historical data. Time Machine will keep filling the backup disk until it's nearly full, then start deleting the oldest backups to free space as needed ⁶ . With 10 TB available and (let's say) a 2 TB iMac internal disk, you could accumulate many months or even years of backups before running out. This long retention is fantastic for research: you might retrieve an old experiment result from last year, for example, as long as the backup from that time still exists. It's essentially an archive of your work over time.

However, **not all data is equally useful to version or retain**. In scientific and AI projects, there are often some directories that churn heavily (temporary files, intermediate model checkpoints, cache files, etc.). If these are included in the backup, they might consume space without real benefit. For example, if you have a cache folder that writes gigabytes and then overwrites them, Time Machine will diligently back up each change, but you likely will never need to restore an old cache state. Apple's guidance is to consider excluding such files: *"find large and frequently updated documents... [and] exclude such files from your Time Machine backup"* ²³ . You can do this by going to Time Machine settings > Options > + (add exclusion). A smart strategy is to **organize large, volatile data onto certain folders**, so you can easily exclude those. Some advanced users even put large data on a separate volume and exclude that entire volume from Time Machine ³⁸ , so that Time Machine focuses on the irreplaceable stuff. For instance, you could store raw AI model training data (if it's reproducible or downloaded from elsewhere) on an external drive or separate partition that isn't backed up, while keeping your unique research results and documents on the main drive (which is backed up). This prevents the backup from ballooning with, say, 500 GB of downloaded ImageNet data that you could re-download if needed. Howard Oakley (Eclectic Light Co.) suggests: if you have sufficient space, it's fine to include everything; but if not, *"to maintain better control of backup storage, you should move VMs and other large files to a separate volume and add that volume to Time Machine's exclude list."* ³⁸ The same principle can apply to large AI datasets or simulation outputs.

One important thing: **never exclude something that you can't afford to lose unless you have another backup of it**. It's easy to get zealous and exclude big folders, but double-check that those truly aren't needed or are backed up elsewhere. For example, if you have a folder of raw experiment data that's huge but only stored on your Mac, you *do* want that backed up (maybe even in multiple ways). On the other hand, a folder of test video files or temporary models might be safely excluded.

Time Machine's *backup coverage* by default is broad and meant to be a safety net. It even backs up things like your user settings, preferences, and emails. After a disaster, restoring from Time Machine can put **everything back in place** exactly as it was – you could boot a new Mac with your Time Machine and get your entire environment (accounts, apps, files) restored. This includes aspects like your development environment, custom scripts, etc., which iCloud or Dropbox wouldn't capture. Apple's documentation confirms you can “*restore your entire system to another Mac*” using Time Machine ³⁹, which underscores its role as a full-system backup (minus the base OS files which are easily reinstalled). For an expert user, this means minimal downtime after a crash – just restore from backup and continue working with all your tools and data intact.

Given that your target audience here is *expert* and this is for scientific/AI usage, a few extra analytical points to note:

- **Data Integrity:** APFS snapshots help ensure that the backup captures a consistent point-in-time image of your data, even if the system is active. For example, if a file is being modified during backup, APFS can use copy-on-write to snapshot the state at start, so Time Machine sees a stable version. This is generally reliable for most files, though for databases or VMs you might still prefer to pause them during backup windows or use their internal backup/export features to get a consistent state. Time Machine is not application-aware (no specific hooks into, say, MySQL or your machine learning processes), so consistency is at a file level. Typically, this is not an issue for things like code, documents, static data, etc. Just keep it in mind if you're doing something like backing up a live database or a virtual machine – those might be better handled with specialized backup procedures or at least be quiesced.
- **Verification and Maintenance:** Apple doesn't provide a user-facing verification tool for Time Machine backups (it periodically checks them silently). As an expert, you might occasionally want to ensure the backup is in good shape. You can use the command-line tool `tmutil` to verify backups or simply try a test restore of some files. Also, monitor that Time Machine is actually running hourly – Ventura's System Settings > Time Machine will show the time of last backup and any errors. Thankfully, issues like the infamous “Time Machine must create a new backup” (common in the older HFS+ days due to sparsebundle corruption) are much rarer now with APFS ⁴⁰. Still, keep an eye on it.
- **Performance Tuning:** By default, you likely won't need to tweak anything. But if you ever wanted to speed up backups at the expense of system resources (for instance, you know you'll be away from the computer for a while and want it to finish a huge backup faster), there is a (undocumented) way to disable the macOS throttling (`sudo sysctl debug.lowpri_throttle_enabled=0`) ⁴¹. This makes Time Machine run full throttle. **Use with caution** – it's not persistent (resets on reboot) and if left off, other background processes might also hog I/O. Generally, it's not recommended except in special cases, and Apple doesn't officially support it. A third-party utility “App Tamer” even provides a toggle for accelerating Time Machine backups by adjusting this setting ⁴². So, the option is there if needed, but in normal usage it's rarely necessary.

Conclusion

Implementing Time Machine on a Mac running macOS Ventura (or newer) with 2017+ hardware – such as your iMac 5K – and pairing it with **RAID1 local storage plus cloud backups** provides a **robust, multi-layered backup strategy**. Let's summarize the key benefits and findings from this configuration:

- **Comprehensive, Automatic Backups:** Time Machine will automatically back up all your important files (apps, documents, research data, etc.) hourly without manual intervention ⁴. It retains multiple versions, allowing easy recovery of accidentally deleted or changed files – a crucial feature when working on scientific papers or AI models where reverting to a prior state can save the day. The hourly/daily/weekly backup scheme ensures even older versions are available (space permitting) ⁶.
- **APFS and Performance on Ventura:** With APFS-based backups, Time Machine on Ventura operates more efficiently and reliably than older macOS versions. Backups are faster and take less space due to block-level differences and compression, as evidenced by Apple's claims and independent benchmarks ³. The initial backup may still be time-consuming, but subsequent incremental backups are much quicker since only changed data is copied ⁸. The macOS scheduler intelligently throttles backup tasks to run in the background, so even on a 2017 iMac you can continue working (including heavy tasks like AI training) while Time Machine runs, with negligible impact on foreground performance ²² ⁴³.
- **RAID1 Mirrored Backup Drive – Data Redundancy:** Using a RAID1 external drive for Time Machine provides **redundancy** for the backup itself. If one of the drives in the mirror fails (a not uncommon event for large drives over time), your backup remains accessible on the other drive. This greatly reduces the risk of losing your only backup due to a drive failure. It aligns with best practices of having multiple copies of data on different devices ²⁶. Keep in mind that while RAID1 protects against hardware failure, it does not replace having an off-site copy – which is why the cloud component is so important. Performance-wise, the mirror does not hinder backup operations noticeably, and modern interfaces (USB 3.1/TB3) and drives can handle the throughput easily. The system treats the RAID as one volume, and thanks to APFS compatibility, Time Machine works seamlessly with it ¹⁴. Also, by encrypting the Time Machine volume, you ensure the backup data is secure from unauthorized access, which is vital for sensitive research data.
- **Cloud Backups – Off-site Protection and Sync:** By incorporating **iCloud and/or third-party cloud services** (like Google Drive, Dropbox), you add an off-site backup layer. This means your most critical data is stored in a secure data center as well, guarding against site-wide disasters (fire, theft, etc.) that could strike both your Mac and local backup. iCloud seamlessly keeps things like your documents, photos, and app data in sync and accessible on other devices ³³. Services like Dropbox can continuously sync research folders. The cloud copies also serve as an immediate remote access option – e.g. you can retrieve a file from anywhere without needing the Time Machine drive. The combination of Time Machine + cloud essentially fulfills the 3-2-1 backup strategy (3 copies, 2 media, 1 off-site) which is considered a gold standard for data protection ³⁶ ²⁶. Do note that cloud services have storage limits and costs, so you might not store *everything* in the cloud (especially large datasets), but even a partial off-site backup of critical items greatly improves your overall data safety net. And unlike Time Machine (which is archival), cloud sync is real-time or very frequent, providing an up-to-the-minute secondary copy for your most active files.

- **Focused on Data Integrity and Recovery:** Our emphasis has been on the **application and benefits** of this backup approach, rather than Apple-specific hardware features. In practical terms, if a file is corrupted or deleted during your AI experiments, you have multiple ways to get it back: Time Machine can pull yesterday's version from the local backup ¹², or iCloud/Dropbox might have a copy if it was synced. If your entire iMac fails, you can restore a replacement Mac from the Time Machine backup (bringing back all your configurations and files) ³⁹, and meanwhile you haven't lost productivity because essential files are in the cloud as well. This layered defense is essential when dealing with valuable scientific data and long-running AI training results that would be costly or impossible to reproduce. By focusing on the **analysis** of the backup performance and coverage, we can conclude that this setup offers a strong balance between speed, reliability, and depth of coverage.

In summary, using Time Machine on macOS Ventura with a 2017+ Mac provides an **efficient, reliable backup system** that, when paired with RAID mirroring and cloud backups, covers all bases: local quick recovery, protection against hardware failure, and protection against catastrophic loss. The performance is generally very good (especially with APFS enhancements), and any speed trade-offs (like throttling or HDD vs SSD limits) are deliberate choices to favor user experience and long-term disk health ²⁰. For an expert user handling scientific papers or AI training material, this approach means you can work with confidence – your data is continuously being safeguarded in the background, with multiple redundancies. The benefit is not just in worst-case scenarios, but also in day-to-day convenience: you can retrieve previous versions of your datasets or code, set up new machines faster, and avoid costly setbacks from data loss. In essence, Time Machine combined with RAID and cloud backup provides a **comprehensive insurance policy** for your research data and workflow, allowing you to focus on your research rather than worrying about data loss.

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